Data Driven Analysis Of University Past Papers & Lecture Content For Curriculum Optimization

25-26J-089

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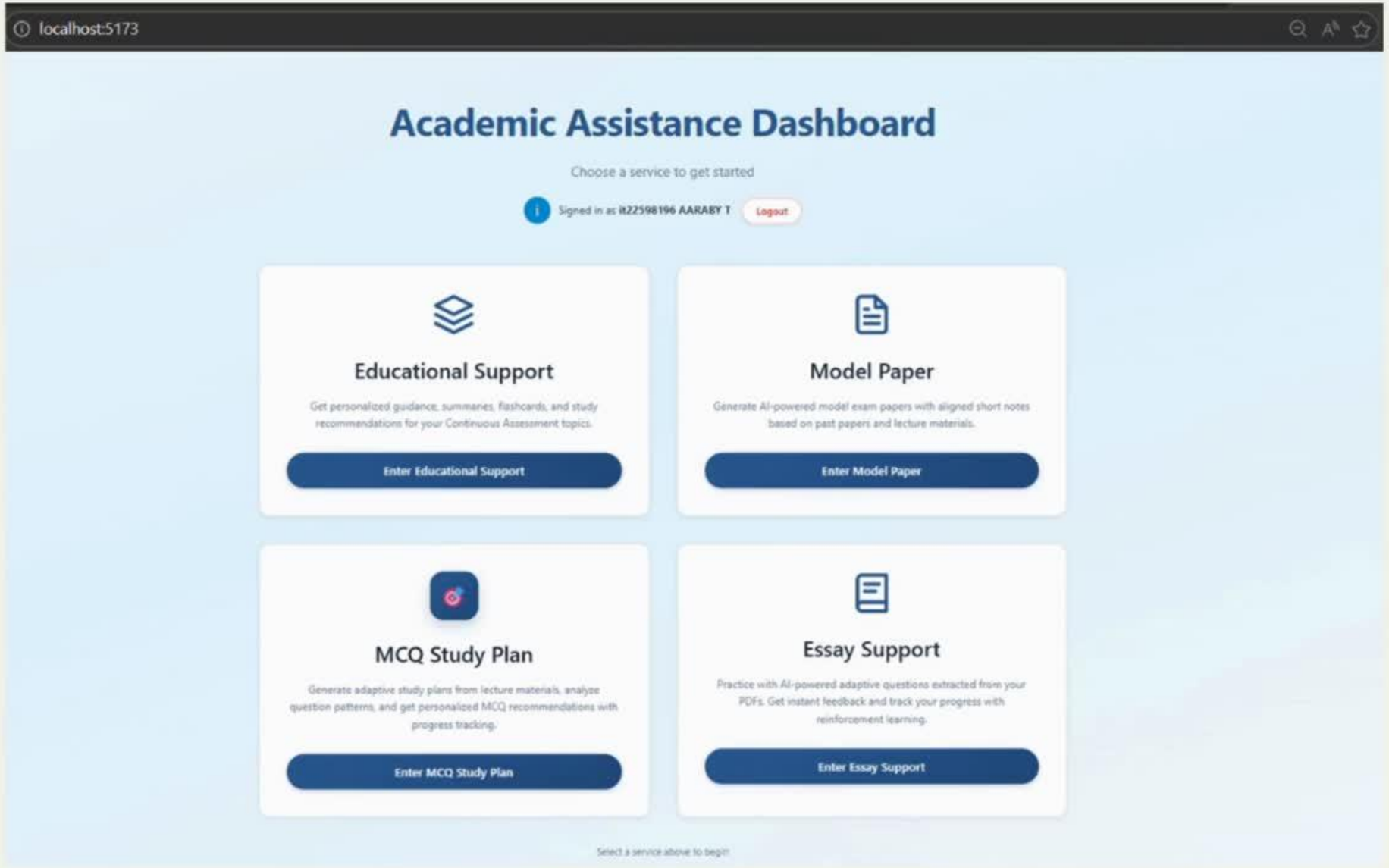
FACULTY OF COMPUTING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Academic Assistance Dashboard
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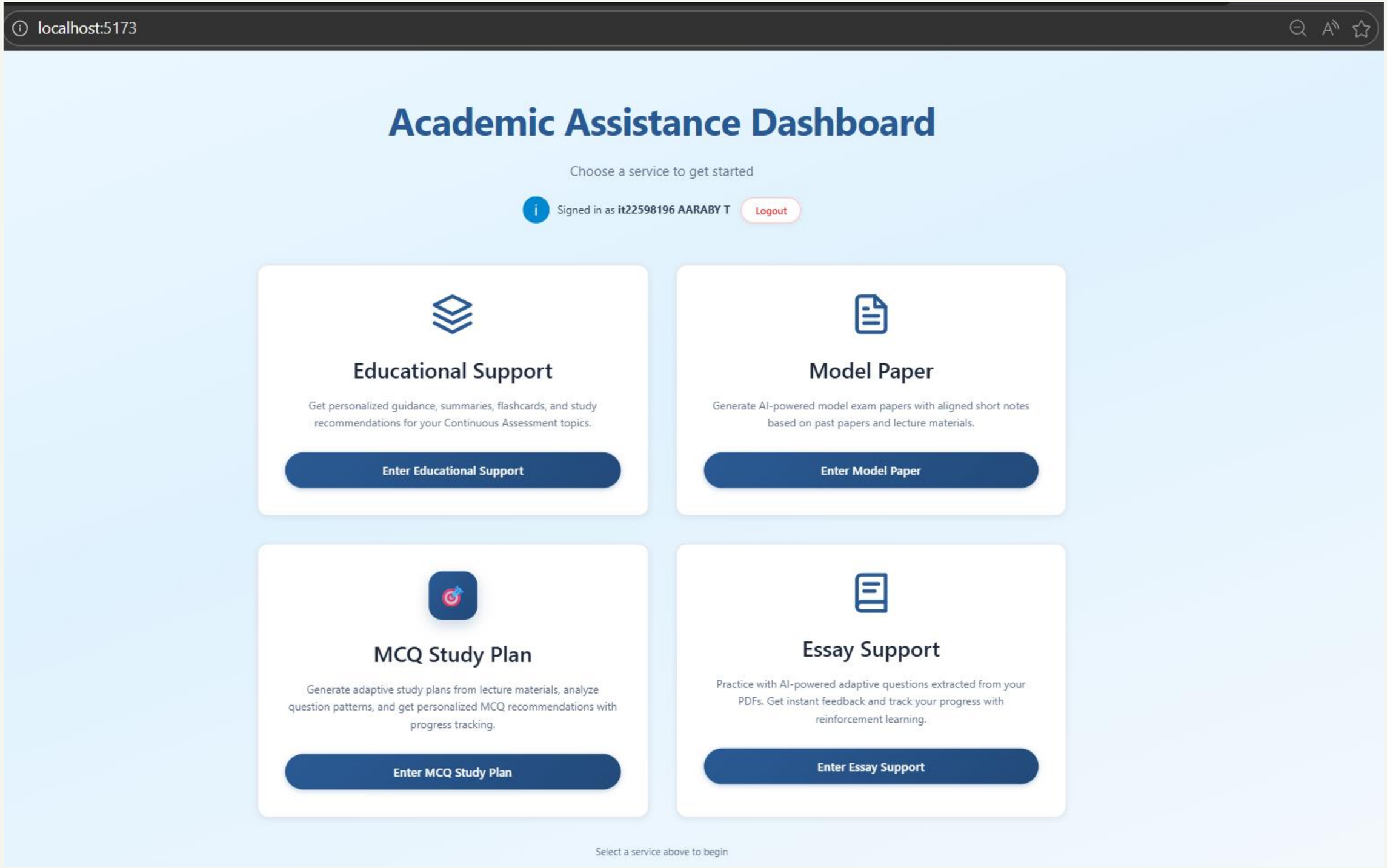
Educational Support
Get personalized guidance, summaries, flashcards, and study recommendations for your Continuous Assessment topics.
Enter Educational Support
Model Paper
Generate AI-powered model exam papers with aligned short notes based on past papers and lecture materials.
Enter Model Paper
MCQ Study Plan
Generate adaptive study plans from lecture materials, analyze question patterns, and get personalized MCQ recommendations with progress tracking.
Enter MCQ Study Plan
Essay Support
Practice with AI-powered adaptive questions extracted from your PDFs. Get instant feedback and track your progress with reinforcement learning.
Enter Essay Support
Select a service above to begin

02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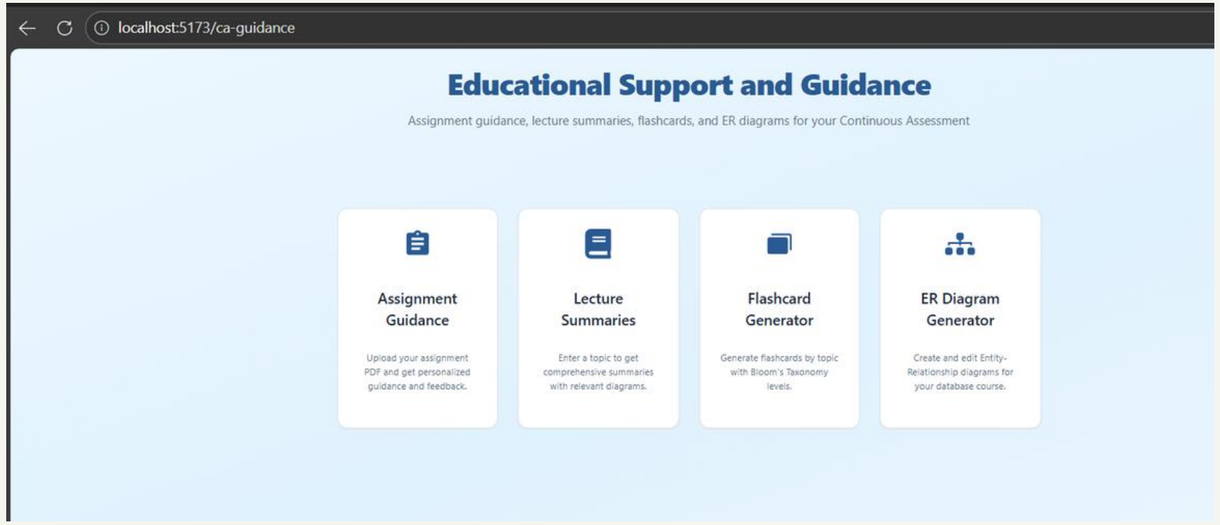
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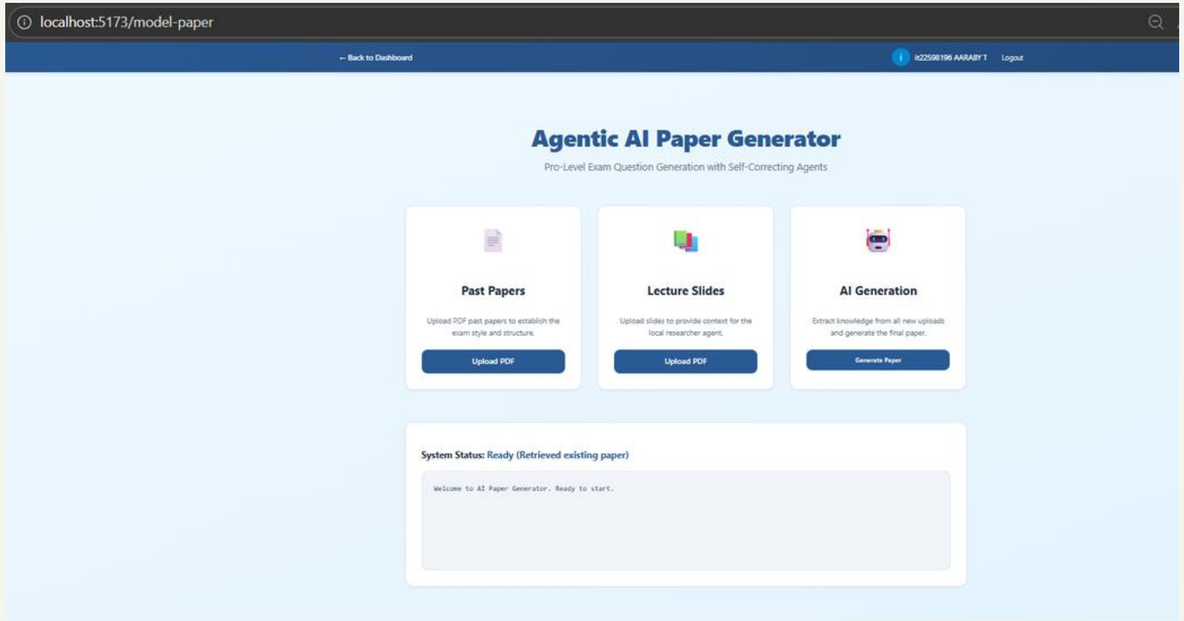
Educational Support and Guidance
Assignment guidance, lecture summaries, flashcards, and ER diagrams for your Continuous Assessment
Assignment Guidance
Upload your assignment PDF and get personalized guidance and feedback.
Lecture Summaries
Enter a topic to get comprehensive summaries with relevant diagrams.
Flashcard Generator
Generate flashcards by topic with Bloom's Taxonomy levels.
ER Diagram Generator
Create and edit Entity-Relationship diagrams for your database course.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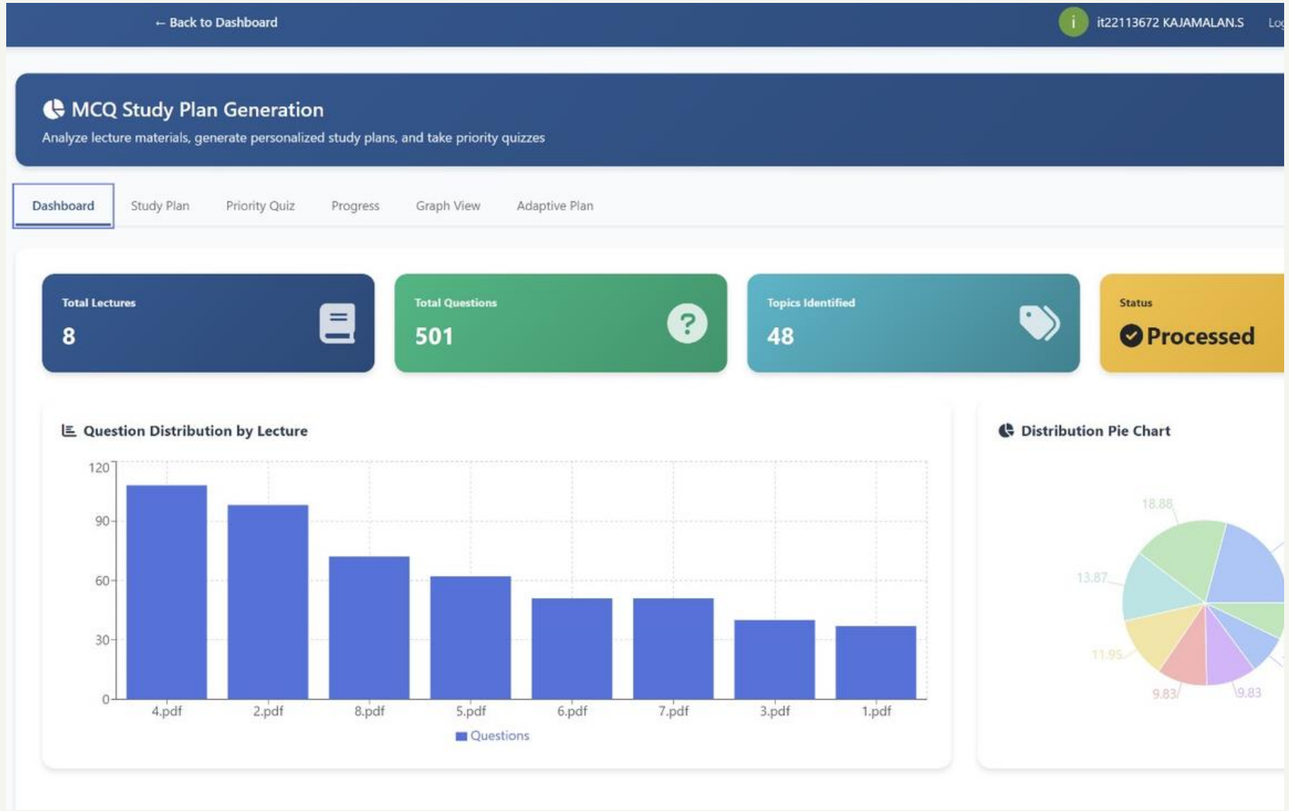
Agentic AI Paper Generator
Pro-Level Exam Question Generation with Self-Correcting Agents
Past Papers
Upload PDF past papers to establish the exam style and structure.
Upload PDF
Lecture Slides
Upload slides to provide context for the local researcher agent.
Upload PDF
AI Generation
Extract knowledge from all new uploads and generate the final paper.
Generate Paper
System Status: Ready (Retrieved existing paper)
Welcome to AI Paper Generator. Ready to start.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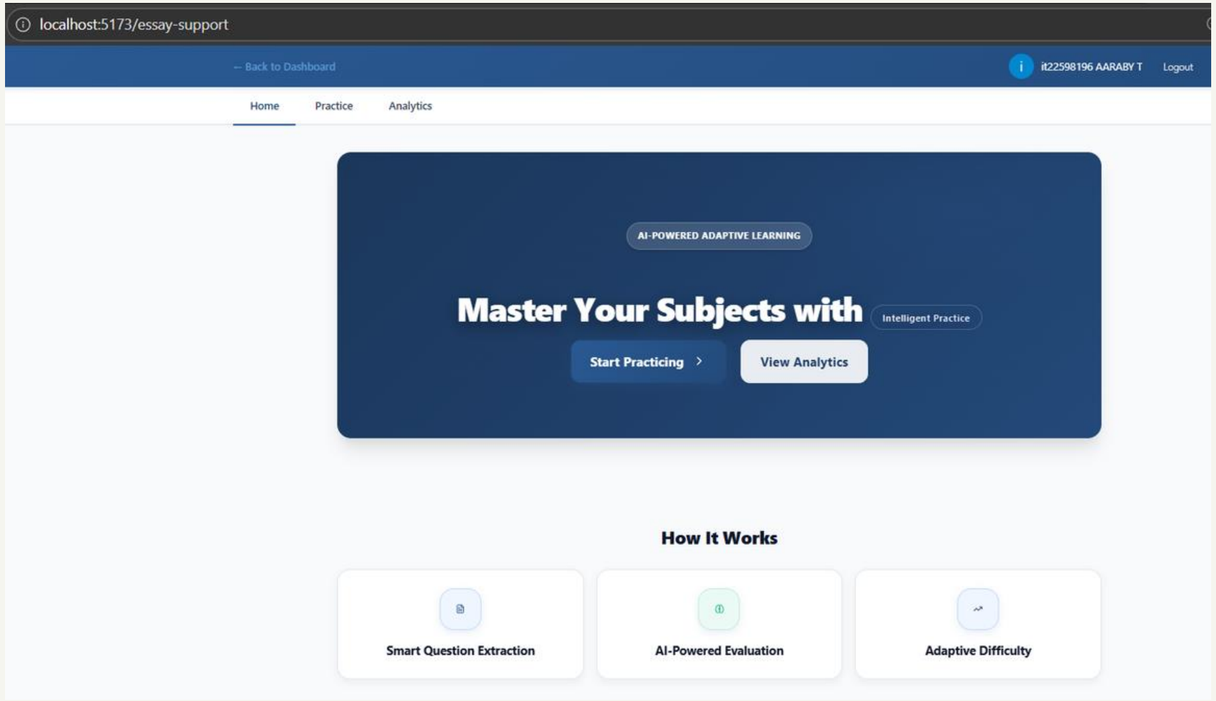
it22113672 KAJAMALAN.S
MCQ Study Plan Generation
Analyze lecture materials, generate personalized study plans, and take priority quizzes
Dashboard
Study Plan
Priority Quiz
Progress
Graph View
Adaptive Plan
Total Lectures
8
Total Questions
501
Topics Identified
48
Status
Processed
Question Distribution by Lecture
120
90
60
30
0
4.pdf
2.pdf
8.pdf
5.pdf
6.pdf
7.pdf
3.pdf
1.pdf
Questions
Distribution Pie Chart
18.88
13.87
11.95
9.83
9.83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Practice
Analytics
AI-POWERED ADAPTIVE LEARNING
Master Your Subjects with
Intelligent Practice
Start Practicing
View Analytics
How It Works
Smart Question Extraction
AI-Powered Evaluation
Adaptive Difficulty

Problem

- Curriculum and exam misalignment in universities
- Students focus on less important topics and miss frequently tested topics
- Manual analysis of past papers and lecture materials
- Lack of analytical tools in Learning Management Systems.
- Generic revision materials without personalized study support

Motivation

- Use data-driven analysis to identify important exam topics
- Provide personalized study resources for students
- Improve learning efficiency and exam preparation
- Develop an AI-powered academic support system

Research Gap

- Existing educational tools mainly support general content learning, but they do not strongly support exam oriented preparation.
- Most current systems do not analyze lecture materials, past exam papers, and continuous assessments together.
- Students often cannot identify high priority topics, question patterns, and difficulty trends from historical exam data.
- There is a lack of AI-driven, personalized academic support systems that provide targeted revision materials, adaptive feedback, and individualized study plans.

Objectives

- **Objective 01**

Develop an AI-based system to analyze past exam papers and lecture slides to generate model exam papers and aligned short notes.

- **Objective 02**

Develop a syllabus-aware learning support system that provides summaries, flashcards, audio explanations, and CA guidance.

- **Objective 03**

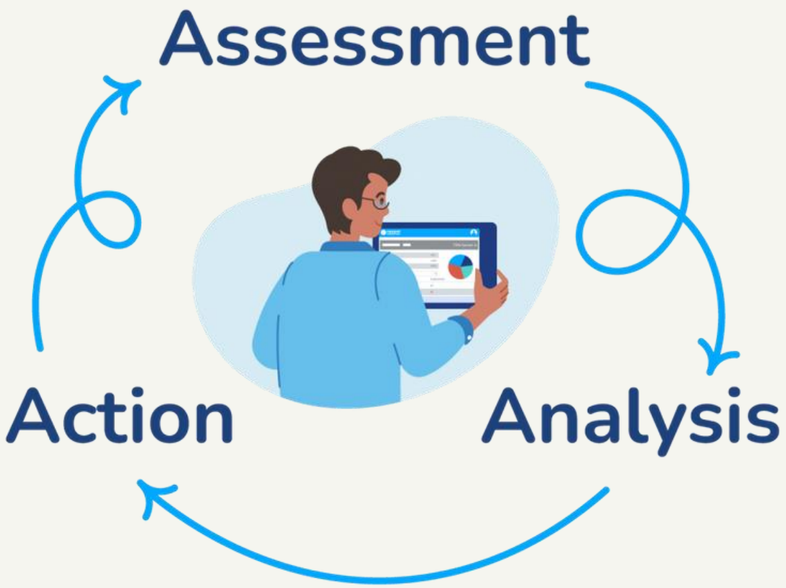
Develop an AI-driven evaluation system to predict question difficulty, provide feedback, and identify important exam topics.

- **Objective 04**

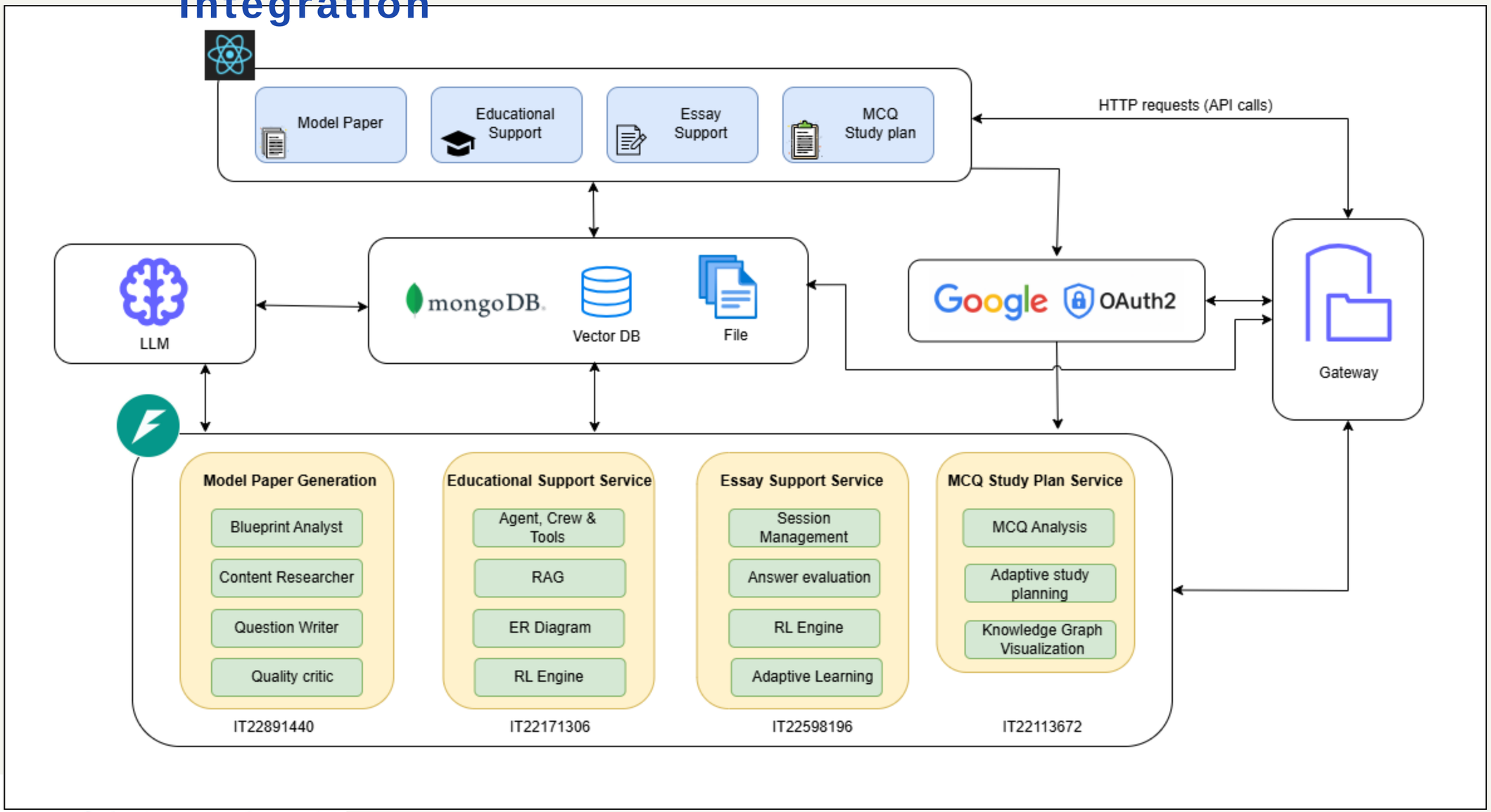
Develop an NLP-based personalization system that classifies educational content and generates personalized MCQ study plans.

Solution Overview

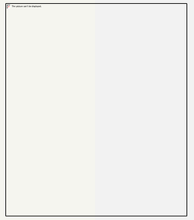
- We built an AI-powered educational support framework that uses NLP, transformer models, semantic analysis, and learning analytics to generate personalized, exam-oriented study resources for university students.



System Architecture & Integration



COMMERCIALIZATION



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